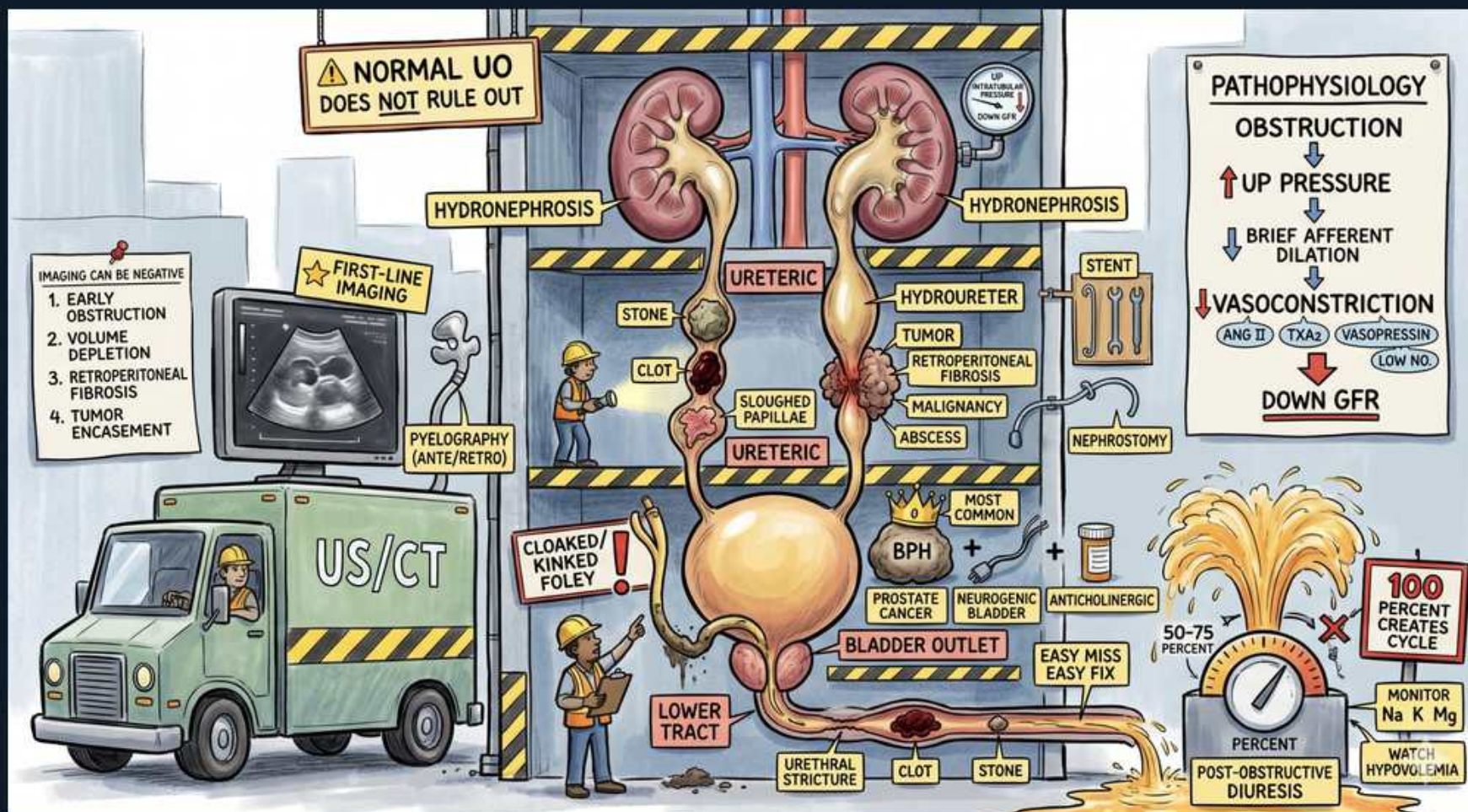


AKI — Postrenal Obstruction



1. Hydronephrosis

WHAT YOU SEE

HYDRONEPHROSIS labels on dilated kidneys

WHAT IT ENCODES

In postrenal AKI, urine cannot drain past the obstruction so back-pressure transmits retrograde and dilates the collecting system, producing hydronephrosis (and hydroureter above a ureteral block). Sonographically this is the hallmark finding and grades from mild calyceal blunting to severe with cortical thinning. The longer the obstruction persists, the more irreversible the parenchymal loss — relief within days usually preserves function, whereas weeks of complete obstruction cause permanent atrophy.

BOARD TRAP

Bilateral hydronephrosis does NOT prove the obstruction is at the bladder/urethra in every case, but combined with a distended bladder it points to outlet obstruction — drain with a Foley first before chasing ureteral causes.

2. Bilateral required

WHAT YOU SEE

The central diagram shows BOTH kidneys swollen with retrograde back-pressure (bilateral hydronephrosis) — both must be obstructed to raise creatinine, because one open kidney keeps GFR near-normal.

WHAT IT ENCODES

To raise the serum creatinine, obstruction must involve BOTH kidneys (or a single functioning/solitary kidney), because one unobstructed kidney has enough reserve to maintain near-normal GFR and a normal creatinine. This is why bladder outlet obstruction (BPH, prostate cancer) is the classic cause of postrenal AKI — it blocks both kidneys at once at the level of the bladder neck/urethra.

BOARD TRAP

WRONG to diagnose postrenal AKI from a unilateral ureteral stone in a patient with two functioning kidneys — the contralateral kidney compensates and creatinine stays normal. A rising creatinine with a unilateral stone should make you look for a second process (volume depletion, contrast, the patient has only one kidney, or bilateral stones).

3. US/CT first-line

WHAT YOU SEE

US/CT truck, FIRST-LINE IMAGING star

WHAT IT ENCODES

Renal ultrasound is the first-line test for suspected obstruction: it is fast, radiation-free, and detects hydronephrosis with high sensitivity. Non-contrast CT (CT KUB) is the test of choice for stones (detects radiolucent uric-acid stones that X-ray misses) and for retroperitoneal masses/fibrosis. Bladder scan/post-void residual >300 mL suggests outlet obstruction.

BOARD TRAP

Do not order CT WITH contrast as the first step in AKI workup — iodinated contrast adds nephrotoxic risk to an already injured kidney, and non-contrast CT or ultrasound answers the obstruction question without it.

4. Normal US doesn't exclude

WHAT YOU SEE

NORMAL UO DOES NOT RULE OUT banner; imaging-can-be-negative box

WHAT IT ENCODES

A normal ultrasound does NOT exclude obstruction. Hydronephrosis can be absent in: (1) very early/acute obstruction before the system has dilated, (2) volume-depleted/oliguric patients with low urine output to distend the system, and (3) encasement by retroperitoneal fibrosis or tumor that traps the ureters and prevents dilation. If clinical suspicion is high, proceed to CT or retrograde pyelography.

BOARD TRAP

WRONG answer is 'normal ultrasound rules out obstruction.' The classic vignette is a patient with metastatic/retroperitoneal malignancy whose ureters are encased — the kidneys can't dilate, so US looks normal despite complete bilateral obstruction. Get a CT.

5. Up pressure → low GFR

WHAT YOU SEE

PATHOPHYSIOLOGY: up pressure, brief afferent dilation, then vasoconstriction, DOWN GFR

WHAT IT ENCODES

Obstruction raises intratubular (Bowman space) hydrostatic pressure, which opposes glomerular filtration and lowers GFR. Renal blood flow has a biphasic response: a brief early afferent vasodilation (prostaglandin/NO-mediated, raising flow for hours) followed by sustained afferent vasoconstriction (thromboxane A2, angiotensin II, vasopressin) that reduces both flow and GFR. Prolonged obstruction also impairs concentrating ability and can cause type IV (hyperkalemic, distal) RTA.

BOARD TRAP

Don't assume creatinine returns to normal the instant you relieve obstruction — recovery lags, and the longer the obstruction, the less complete the recovery.

6. Ureteric stone

WHAT YOU SEE

STONE / CLOT on ureter, URETERIC labels

WHAT IT ENCODES

Intraluminal upper-tract causes include ureteral stones, blood clots (after biopsy/tumor/anticoagulation), sloughed renal papillae (papillary necrosis from diabetes, sickle cell, analgesic/NSAID abuse), and urothelial tumor. Stones lodge at the three anatomic narrowings: ureteropelvic junction, pelvic brim (iliac crossing), and ureterovesical junction. Remember: a single ureteral stone causes AKI only if it is the patient's solitary or sole functioning kidney.

BOARD TRAP

Sloughed papilla can mimic a stone on imaging — look for the predisposing condition (Diabetes, Sickle cell, Analgesics/NSAIDs, Pyelonephritis — mnemonic 'DSAP' / POSTCARDS) to distinguish papillary necrosis from a true calculus.

7. Retroperitoneal fibrosis

WHAT YOU SEE

Retroperitoneal fibrosis, malignancy, abscess labels

WHAT IT ENCODES

Extrinsic (outside-the-wall) ureteral compression causes obstruction without an intraluminal lesion: retroperitoneal fibrosis (idiopathic/Ormond disease, IgG4-related, or drug-induced — methysergide, ergots), malignancy (cervical, prostate, colorectal, lymphoma, retroperitoneal sarcoma), and abscess/hematoma. Retroperitoneal fibrosis classically encases the ureters and pulls them medially.

BOARD TRAP

Because the ureters are encased and cannot dilate, retroperitoneal fibrosis is a top cause of OBSTRUCTION WITH NORMAL/MINIMAL HYDRONEPHROSIS on ultrasound — the discriminator over 'obstruction always shows hydronephrosis.' Workup needs CT/MRI; IgG4-related disease is steroid-responsive.

8. BPH

WHAT YOU SEE

BPH, prostate cancer, neurogenic bladder at bladder outlet

WHAT IT ENCODES

Bladder outlet obstruction is the SINGLE MOST COMMON cause of postrenal AKI, especially in older men. Causes: benign prostatic hyperplasia (most common), prostate cancer, bladder/cervical cancer, and neurogenic bladder (diabetes, MS, spinal cord injury). Because it blocks both kidneys simultaneously at the outlet, it reliably raises creatinine. First management step is bladder decompression with a urinary catheter.

BOARD TRAP

WRONG to jump to ureteral stenting or nephrostomy in an older man with a distended bladder and AKI — a simple Foley catheter relieves outlet obstruction and is the immediate, definitive first step. Always palpate/scan the bladder.

9. Anticholinergic

WHAT YOU SEE

Anticholinergic at bladder outlet

WHAT IT ENCODES

Anticholinergic drugs (antihistamines like diphenhydramine, TCAs, oxybutynin, antipsychotics, antiparkinsonian agents) and opioids cause FUNCTIONAL urinary retention by impairing detrusor contraction — a reversible cause of postrenal AKI that needs no surgery, just stopping the drug and draining the bladder. Sympathomimetics (decongestants) worsen retention via bladder-neck contraction, especially atop BPH.

BOARD TRAP

In an elderly patient with new urinary retention/AKI, review the medication list FIRST — a recently started anticholinergic or opioid is a reversible cause that is easily missed when you anchor on BPH.

10. Blocked Foley

WHAT YOU SEE

CLOAKED/KINKED FOLEY, 'easy miss easy fix'

WHAT IT ENCODES

A kinked, clamped, or clot-obstructed Foley catheter is the classic 'easy miss, easy fix' cause of postrenal AKI in a hospitalized patient: the catheter is in place so obstruction seems excluded, yet urine isn't draining. The bedside maneuver is to flush/irrigate or exchange the catheter — if output resumes, the catheter was the problem.

BOARD TRAP

Don't assume a patient 'can't be obstructed because they have a Foley.' A blocked catheter perfectly mimics anuria from intrinsic AKI — flush it before launching an expensive renal workup.

11. Urethral stricture

WHAT YOU SEE

URETHRAL STRICTURE, clot, stone at lower tract

WHAT IT ENCODES

Lower-tract (urethral) obstruction includes urethral stricture (from prior instrumentation, trauma, STIs/gonococcal urethritis, or radiation), urethral stones, clots, and posterior urethral valves in boys. Strictures present with a weak stream and difficult catheter passage. Suprapubic catheterization may be needed when a urethral catheter cannot be passed.

BOARD TRAP

Forcing a Foley past a tight urethral stricture can create a false passage — if catheterization fails, call urology for guidewire/dilation or suprapubic drainage rather than repeated forceful attempts.

12. Stent / nephrostomy

WHAT YOU SEE

STENT and NEPHROSTOMY relieving obstruction

WHAT IT ENCODES

Definitive treatment is relieving the obstruction at its level: Foley/suprapubic catheter for outlet obstruction; retrograde ureteral stent (cystoscopic) or percutaneous nephrostomy tube for ureteral obstruction. Nephrostomy is preferred when retrograde stenting fails or in infected/obstructed systems (pyonephrosis) requiring urgent drainage. Decompression both restores GFR and treats the urosepsis risk.

BOARD TRAP

An obstructed, infected kidney (pyonephrosis/obstructive pyelonephritis with sepsis) is a urologic EMERGENCY — antibiotics alone are insufficient; the system must be drained emergently by stent or nephrostomy or the patient progresses to septic shock.

13. Post-obstructive diuresis

WHAT YOU SEE

50-75 PERCENT post-obstructive diuresis gauge

WHAT IT ENCODES

After relief of bilateral/chronic obstruction, post-obstructive diuresis develops in roughly 50–75% of patients: a transient appropriate diuresis (excreting retained urea, Na, and water) that can become pathologic with massive polyuria, hypovolemia, and electrolyte wasting (hypokalemia, hyponatremia, hypomagnesemia, hypophosphatemia). Monitor urine output, vitals, and electrolytes; replace fluids if output is excessive (commonly replace ~half of urine output with hypotonic fluid).

BOARD TRAP

WRONG to reflexively match urine output mL-for-mL with IV fluids — aggressive 1:1 replacement perpetuates the diuresis. Replace partially (about 50–75% of losses) and let the kidney down-regulate; most post-obstructive diuresis is physiologic and self-limited.

14. Cr clearance cycle

WHAT YOU SEE

100 PERCENT CREATININE CYCLE gauge on the right

WHAT IT ENCODES

Postrenal AKI is one of the most reversible forms of AKI: because the parenchyma is initially intact, prompt relief of obstruction can restore creatinine toward 100% of baseline. The probability of full recovery falls with duration and completeness of obstruction — relief within a week usually recovers fully, partial obstruction recovers better than complete, and chronic obstruction leaves fixed loss from tubulointerstitial fibrosis.

BOARD TRAP

Don't label slow creatinine recovery after relief as 'failed treatment' — recovery is gradual over days, and a baseline that doesn't fully return reflects pre-existing chronic obstructive damage, not a missed second diagnosis.